Numerical Computing

Roots of Equations

- **Introduction & Python for Numerical Computing**
- **Computer Arithmetic & Sources of Error**
- **Roots of Equations**
  - Bisection method
  - Fixed-point iteration
  - Newton–Raphson method
  - Secant method
  - Implementations in Python
- Interpolation & Approximation
- Numerical Integration
- Numerical Differentiation & ODEs
- Numerical Linear Algebra
- Project

Intermediate Value Theorem

Intermediate Value Theorem (IVT)

- **Statement of IVT:**

  If $f(x)$ is **continuous** on $[a, b]$ and
  $f(a)$ and $f(b)$ have **opposite signs**,
  then there exists $p \in (a, b)$ such that

  $$f(p) = 0$$

- **Interpretation:**

  The function **must cross the x-axis** somewhere in the interval.

- **Key Insight:**
  - IVT guarantees **existence** of at least one root.
  - But it does **not give the exact location** of the root.

Bisection Method

Bisection Method (Using IVT)

- Start with $[a, b]$ where $f(a) \cdot f(b) < 0$
- **Iteration Step:**
  1. Compute midpoint $c = (a + b)/2$
  2. Evaluate $f(c)$
  3. Choose subinterval where **sign change** occurs

     → new interval becomes $[a, c]$ or $[c, b]$
- Repeat until interval is sufficiently small.
- After n iterations:

$$\text{Interval length} = \frac{b - a}{2^n}$$

- Midpoint c gives an approximation to the root.

Roots of Equations